

Evaluate the following integrals:

$$1. \int \frac{2}{x-3} dx = 2 \ln |x-3| + C$$

$$2. \int \frac{2}{x-3} + \frac{5}{(x-3)^2} - \frac{1}{(x-3)^3} dx = 2 \ln |x-3| - 5(x-3)^{-1} - \frac{(x-3)^{-2}}{-2} + C$$

$$3. \int \cos^2 x \sin x dx = -\frac{\cos^3 x}{3} + C$$

$$4. \int \frac{5}{\sqrt{x^2+4}} dx = 5 \ln \left| \frac{\sqrt{x^2+4}}{2} + \frac{x}{2} \right| + C$$

$$5. \int x e^{2x} dx = \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C$$

$$6. \int x e^{x^2} dx = \frac{1}{2} e^{x^2} + C$$

$$7. \int \sin^2(3x) dx = \frac{1}{2} \left(x - \frac{1}{6} \sin(6x) \right) + C$$

$$8. \int \frac{3}{x^2+4x} dx = \frac{3}{4} \ln |x| + \frac{-3}{4} \ln |x+4| + C$$

$$9. \int 2x \cos(3x) dx = \frac{2}{3} x \sin(3x) + \frac{2}{9} \cos(3x) + C$$

$$10. \int \sqrt{5-x^2} dx = \frac{5}{2} \left(\sin^{-1} \left(\frac{x}{\sqrt{5}} \right) + \frac{x}{\sqrt{5}} \frac{\sqrt{5-x^2}}{\sqrt{5}} \right) + C$$

$$11. \int \frac{x}{x^2-25} dx = \frac{1}{2} \ln |x^2-25| + C$$

$$12. \int e^{2x} \sin(5x) dx = \frac{25}{29} \left(\frac{-1}{5} e^{2x} \cos(5x) + \frac{2}{25} e^{2x} \sin(5x) \right) + C$$

$$13. \int \frac{2x}{x^2 + 6x + 5} dx = \int \frac{5}{2(x+5)} - \frac{1}{2(x+1)} dx = \frac{5}{2} \ln|x+5| - \frac{1}{2} \ln|x+1| + C$$

$$14. \int \frac{x-1}{x^2 + 6x + 9} dx = 4(x+3)^{-1} + \ln|x+3| + C$$

$$15. \int \frac{1}{x^2 + 6x + 14} dx = \frac{1}{\sqrt{5}} \tan^{-1} \left(\frac{x+3}{\sqrt{5}} \right) + C$$

$$16. \int \frac{1}{\sqrt{x^2 + 6x + 14}} dx = \ln \left| \frac{\sqrt{5 + (x+3)^2}}{\sqrt{5}} + \frac{x+3}{\sqrt{5}} \right| + C$$

$$17. \int \frac{2x+3}{x^2 + 6x + 14} dx = \ln|(x+3)^2 + 5| - \frac{3}{\sqrt{5}} \tan^{-1} \left(\frac{x+3}{\sqrt{5}} \right) + C$$

$$18. \int \frac{3 \sin x dx}{\cos^2 x + 9} = -\tan^{-1} \left(\frac{\cos x}{3} \right) + C$$

$$19. \int x^4 \sin(2x) dx = -\frac{1}{2} x^4 \cos(2x) + x^3 \sin(2x) + \frac{12}{8} x^2 \cos(2x) - \frac{24}{16} x \sin(2x) - \frac{24}{32} \cos(2x) + C$$

$$20. \int \frac{\sin(\ln 3x)}{2x} dx = -\frac{1}{2} \cos(\ln(3x)) + C$$

$$21. \int \frac{4x+2}{x^2 - x} dx = 6 \ln|x-1| - 2 \ln|x| + C$$

$$22. \int \frac{6x^2 - 7x}{3x - 2} dx = x^2 - x - \frac{2}{3} \ln|3x - 2| + C$$